THE MASTERMIND COACHING CLASSES
CBSE / RBSE • NURSERY – CLASS XII • JAIPUR
Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 9: **The Amazing World of Solutes, Solvents and Solutions** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** homogeneous mixture	Uniform.
2	**(c)** hot water	Heat helps.
3	**(b)** mass of solute + mass of solvent	Total.
4	**(b)** solute	Dissolved substance.
5	**(b)** faster	Mixing.
6	**(c)** 100 g	Sum.
7	**(d)** solvent	Dissolver.
8	**(d)** faster	Surface.
9	**(b)** temperature	Heat.
10	**(d)** solution	True solution.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Homogeneous mixture of solute and solvent (2).	2
12	Heterogeneous mixture whose particles settle (2).	2
13	Dissolves many substances (2).	2
14	Substance that dissolves in a solvent (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Sugar and water (1); CO2 and water (1); Iodine and alcohol (1).	1+1+1
16	Solution = 50 g (1); 10/50 x 100 (1); 20 per cent (1).	1+1+1
17	Solid in liquid (1); Gas in liquid (1); Gas in gas (1).	1+1+1
	OR: Homogeneous (1); Transparent (1); Particles don't settle (1).	1+1+1
18	15 + 85 (1); 100 g (1); Percentage 15 (1).	1+1+1
	OR: Transport of nutrients (1); Removal of wastes (1); Body reactions (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Solute (1); Solvent (1); Sugar-water (1); CO2-water (1); Iodine-alcohol (1).	1+1+1+1+1
	OR: Dissolves many solids (1); Dissolves gases (1); Dissolves liquids (1); Used in life (1); Not all substances (1).	1+1+1+1+1
20	Solution: sugar water (1); Suspension: muddy water (1); Colloid: milk (1); Settling (1); Tyndall effect (1).	1+1+1+1+1
	OR: Solid-solid: brass (1); Gas-liquid: soda (1); Gas-gas: air (1); Liquid-liquid: vinegar (1); Solid-liquid: salt water (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 9: **The Amazing World of Solutes, Solvents and Solutions** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** concentrated	More solute.
2	**(b)** temperature rise	Gas leaves.
3	**(c)** suspension	Settles.
4	**(b)** dilute	Less solute.
5	**(d)** Tyndall effect	Small particles.
6	**(b)** colloid	Light scattering.
7	**(b)** sand from water	Insoluble solid.
8	**(b)** large	Visible.
9	**(d)** water	Dissolves many.
10	**(c)** salt from salt solution	Dissolved solid.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Salt water (1); Sugar water (1).	1+1
12	Stirring (1); Heating (1).	1+1
13	Substance in which a solute dissolves (2).	2
14	Muddy water (1); Chalk in water (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Heterogeneous (1); Particles settle (1); Can be filtered (1).	1+1+1
16	Particle size (1); Settling (1); Filtration (1).	1+1+1
17	Pressure falls (1); CO_2 escapes (1); Gas solubility decreases (1).	1+1+1
	OR: ORS (1); Sherbet (1); Soap solution (1).	1+1+1
18	For respiration (1); Gills take oxygen from water (1); Without it they die (1).	1+1+1
	OR: Milk (1); Fog (1); Butter (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Definition (1); Temperature (1); Nature of solute (1); Nature of solvent (1); Pressure for gas (1).	1+1+1+1+1
	OR: Saturated (1); Unsaturated (1); Test by adding more solute (1); Effect of heating (1); Example (1).	1+1+1+1+1
20	Solution = 100 g (1); 20/100 x 100 (1); 20 per cent (1); Water = 80 g (1); Unit correct (1).	1+1+1+1+1
	OR: Solubility decreases (1); Fish and warm water (1); Soda goes flat (1); Cold keeps gas (1); Pressure effect (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 9: **The Amazing World of Solutes, Solvents and Solutions** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** mud settles	Suspension.
2	**(c)** oil	Immiscible.
3	**(b)** gases	Gas in gas.
4	**(c)** larger amount	Usually.
5	**(d)** cannot dissolve more solute	Full.
6	**(a)** metals	Alloy.
7	**(d)** per cent	Percentage.
8	**(a)** carbon dioxide	Gas.
9	**(a)** dissolved in water	Dissolved gas.
10	**(c)** alcohol	Alcohol.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Evaporation (1); Salt remains (1).	1+1
12	Solution that cannot dissolve any more solute (2).	2
13	Amount of solute that dissolves in 100 g of solvent at a given temperature (2).	2
14	Adding more solvent to a solution (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Stirring (1); Heating (1); Crushing (1).	1+1+1
16	Oxygen (1); Carbon dioxide (1); Nitrogen (1).	1+1+1
17	Dissolve in water (1); Filter sand (1); Evaporate for salt (1).	1+1+1
	OR: 5 per cent of 100 (1); 5/100 x 100 (1); 5 g (1).	1+1+1
18	Particles move faster (1); More solubility (1); Quick mixing (1).	1+1+1
	OR: Solution with less solute (1); Weak tea (1); Add more solvent (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Two glasses of water (1); Add sugar to both (1); Stir one (1); The stirred one dissolves faster (1); Same temperature (1).	1+1+1+1+1
	OR: Boil the solution (1); Steam collects (1); Condenses (1); Water obtained (1); Distillation (1).	1+1+1+1+1
20	Add water (1); Salt dissolves (1); Filter (1); Sand on filter paper (1); Evaporate to get salt (1).	1+1+1+1+1
	OR: Kitchen: sherbet (1); Kitchen: tea (1); Medicine: ORS (1); Lab: reagent (1); Cleaning: soap (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.